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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch : VII Semester B.E. ECE A
Course Code & Title : EC8791 Embedded and Real Time Systems
Name of the Faculty member : Mr.G.Sivakumar

Innovative Practice Description

- **Unit / Topic:** Unit IV / Task Assignment
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 4.2
- **Activity Chosen:** Flipped Classroom
- **Justification:** This allows students to learn at their own pace, encourages them to actively engage with lecture material, frees up actual class time for more effective, creative, and active learning activities, and allows them to be in control of their learning.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
  - ➢ The topic was given to the students learn on their own. Resources like reference book, PPT and videos were already post in canvas. This flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related
  - ➢ In this blended learning approach, face-to-face interaction is mixed with independent study–usually via technology. In a common Flipped Classroom scenario, students might watch pre-recorded videos at home, then come to school to do the homework armed with questions and at least some background knowledge.
  - ➢ The concept behind the flipped classroom is to rethink when students have access to the resources they need most. If the problem is that students need help doing the work rather than being introduced to the new thinking behind the work, then the solution the flipped classroom takes is to reverse that pattern.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 4	PO 9	PO 10	PO 12	PSO 1	PSO 2
CO3	3	2	2	1	1	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 3
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Flipped Classroom activity to discuss about different Task Assignment Algorithms, So Course outcome is mapped at level 1	In Flipped Classroom activity students will make effective presentations about Task Assignment Algorithms, hence Course outcome is mapped at level 1	Knowledge of Task Assignment Algorithms is used to Design, analyze and develop any real time applications in the area of Embedded Applications, So Course outcome is mapped at level 3

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students eagerly participated and enjoyed this activity.
  - ***Benefit of the practice:***
    - Outcome attainment would have increased due to innovative practice over conventional practice.
    - Students easily understand the concepts of Task Assignment Process.
    - Students can answer easily about different types of Task Assignment Algorithms.

- ***Challenges faced in implementation:***
  - Faced issues while make the students to understand about Flipped Class Room Activity rules.
  - Make the students to learn about Task Assignment Algorithms before the start of this activity.

References:

- Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html.

Signature of Faculty Member

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Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch : VII Semester B.E. ECE A
Course Code & Title : EC8791 Embedded and Real Time Systems
Name of the Faculty member : Mr.G.Sivakumar

Innovative Practice Description

- **Unit / Topic:** Unit II / Instruction Set
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 2.1
- **Activity Chosen:** Jigsaw
- **Justification:**
  - ➢ The activity 'JIGSAW' is used to recall the concepts learnt and enhance the learning through this collaboration. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity.
  - ➢ ARM is a popular and efficient instruction set for processors, focusing on simplicity, speed, and low power. so the activity 'JIGSAW' is used to recall the concepts learnt and enhance the learning through this collaboration.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

  - ➢ **Materials for the Activity:**
    The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.
  - ➢ Plan for Implementing this Activity:
    Lessons are classified into subcategories. Each group is assigned a topic, the student learns about his or her topic, preparatory materials are put in CANVAS, and the student presents it to the group. The Class Strength is divided into 8 Groups with 6 members Following that, an expert group is formed with members from each group and discusses their topic for 20 minutes with its members, which has previously been discussed with the home group. After that, students gather into groups divided by topic. Finally, one from each group should summarize the points of the topic Design as a tradeoff to the class 20 Minutes. A member of the expert group presents subjects to members of the home group. It is a cooperative learning strategy that promotes both individual accountability and team goal achievement.

Each of these groups is given a different topic to learn about. Review of points and conclusion of the activity (10 minutes).

- **CO – PO / PSO mapping:**

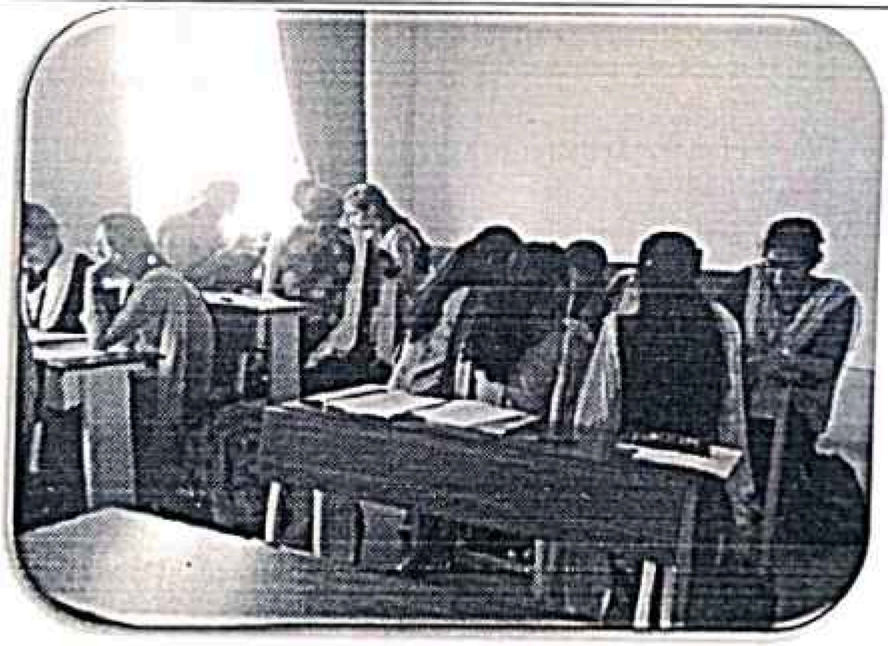
CO	PO 1	PO 2	PO 3	PO 9	PO 10	PO 12	PSO 1	PSO 3
CO3	3	3	3	1	1	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 3
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Jigsaw activity to discuss about various Instruction Sets, So Course outcome is mapped at level 1	In Jigsaw activity students will make effective explanations on various Instruction Sets, hence Course outcome is mapped at level 1	various Instruction Sets used to write code for ARM Processors to develop real time applications Embedded Systems, So Course outcome is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students eagerly participated and enjoyed this activity.
  - ***Benefit of the practice:***
    - Students easily understand the concepts of various Instruction Sets.
    - Students can easily write their own code for any real life applications.
  - ***Challenges faced in implementation:***
    - Lack of preparation: Making the students to learn the materials is the challenging task.
    - Each student has to read the material posted in the course website.
    - Non participation in the activity: All the students should be made involve in the activity
    - Time management: The Jigsaw activity should be able to complete in the planned duration.

References:

- https://www.readingrockets.org/strategies/jigsaw
- https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/

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<u>Innovative Practices Description</u>

Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title:EC8791 Embedded and Real Time Systems
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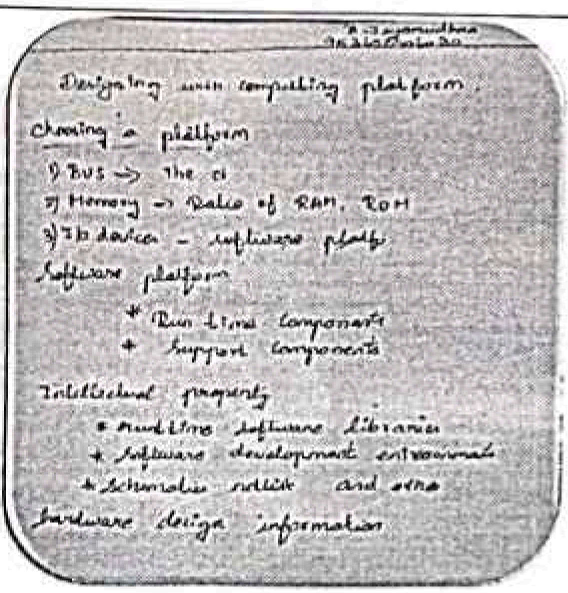
UNIT I	Date: 11.08.2023
Name of the Topic	**Name of the Innovative Practice**
Designing with Computing Platforms	One Minute Paper



Designing with computing platform.
Choosing a platform
1) Bus → The ci
2) Memory → Ratio of RAM, ROM
3) I/O devices – software platf
Software platform
* Run time components
* Support components
Intellectual property
* Run time software libraries
* Software development environment
* Schematics netlist and other
hardware design information

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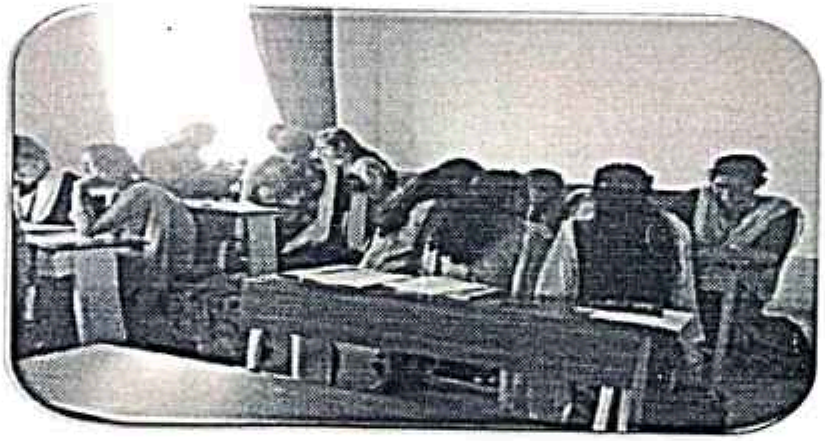
UNIT II	**Date: 22.08.2023**
Name of the Topic	**Name of the Innovative Practice**
Instruction Set	Jig Saw



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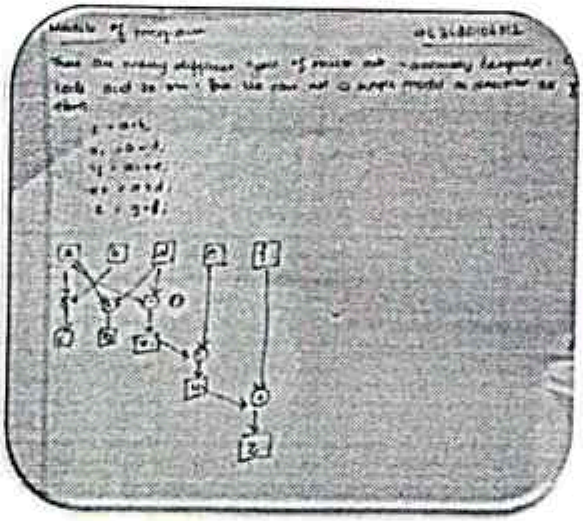
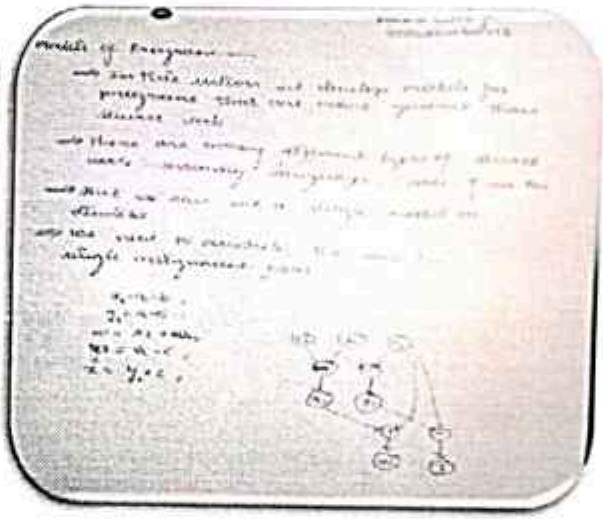
UNIT III	**Date: 19.09.23**
Name of the Topic	**Name of the Innovative Practice**
Models of Programs	Mind Map





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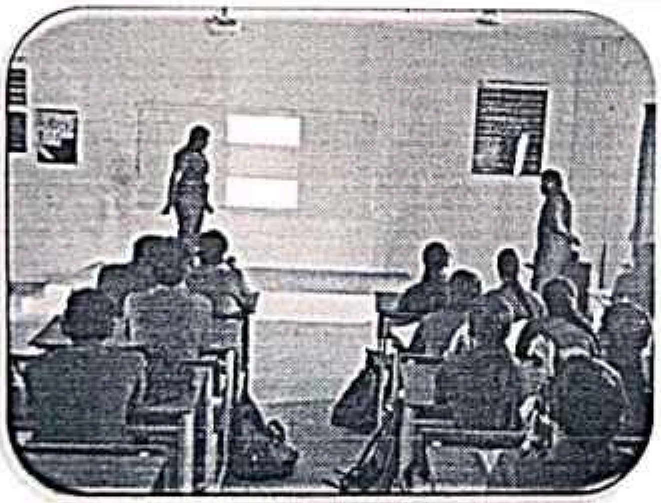
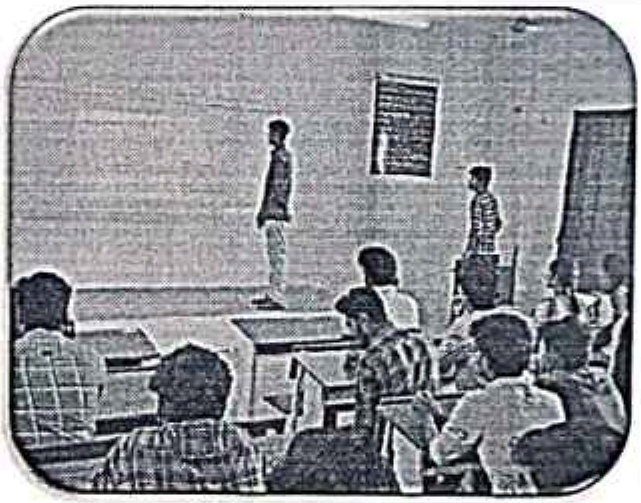
UNIT IV	Date :10.10.2023
Name of the Topic	**Name of the Innovative Practice**
Task Assignment	Flipped Class Room

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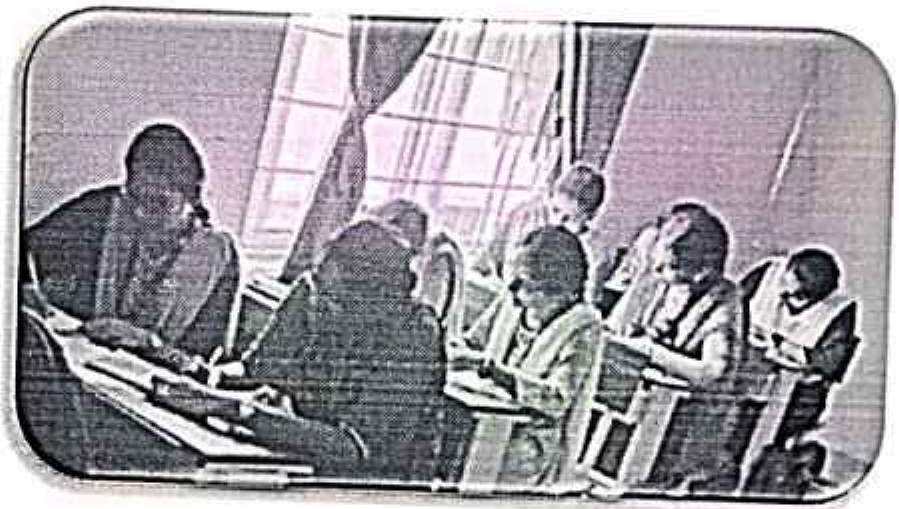
UNIT V	
Name of the Topic	**Name of the Innovative Practice**
Distributed Embedded Systems	Open Book Test



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